

# OpenBioShield — Assay Design Report

Assay ID: 6f753ddc-f0be-4082-8b3a-514dcde4bebc | Generated: 2026-06-09T07:34:17.317267Z | Candidates: 10

## Top Primer Candidates

| # | Forward (5'→3')      | Reverse (5'→3')      | Tm F (°C) | Tm R (°C) | GC F (%) | GC R (%) | Prod (bp) | Cov   | Thermo | AI   | Score  |
|---|----------------------|----------------------|-----------|-----------|----------|----------|-----------|-------|--------|------|--------|
| 1 | TGGACCCCAAATCAGCGAA  | TTGAGTGAGAGCGGTGAACC | 62.9      | 62.9      | 50.0     | 55.0     | 163       | 100.0 | 100.0  | 68.8 | 93.770 |
| 2 | TGGACCCCAAATCAGCGAA  | GGGCCGACGTTGTTTTGATC | 62.9      | 63.0      | 50.0     | 55.0     | 115       | 100.0 | 100.0  | 68.8 | 93.752 |
| 3 | TGGACCCCAAATCAGCGAA  | CCATGTTGAGTGAGAGCGGT | 62.9      | 63.0      | 50.0     | 55.0     | 168       | 100.0 | 100.0  | 68.7 | 93.730 |
| 4 | GGTTCACCGCTCTCACTCAA | TCCACACCATGTTTGCTCCA | 62.9      | 62.8      | 55.0     | 50.0     | 152       | 100.0 | 100.0  | 68.3 | 93.666 |
| 5 | ACCGCTCTCACTCAACATGG | TCCACACCATGTTTGCTCCA | 63.0      | 62.8      | 55.0     | 50.0     | 147       | 100.0 | 100.0  | 68.1 | 93.624 |

## Score Weights

| Component      | Weight |
|----------------|--------|
| Coverage Score | 60%    |
| Thermodynamic  | 20%    |
| AI Efficiency  | 20%    |